Real-Time Credit Card Fraud Detection build with IBM Bob

Install Confluent Kafka, develop & document Credit Card Fraud in real time with IBM Bob

Real-Time Credit Card Fraud Detection with Confluent Kafka

The Challenge:

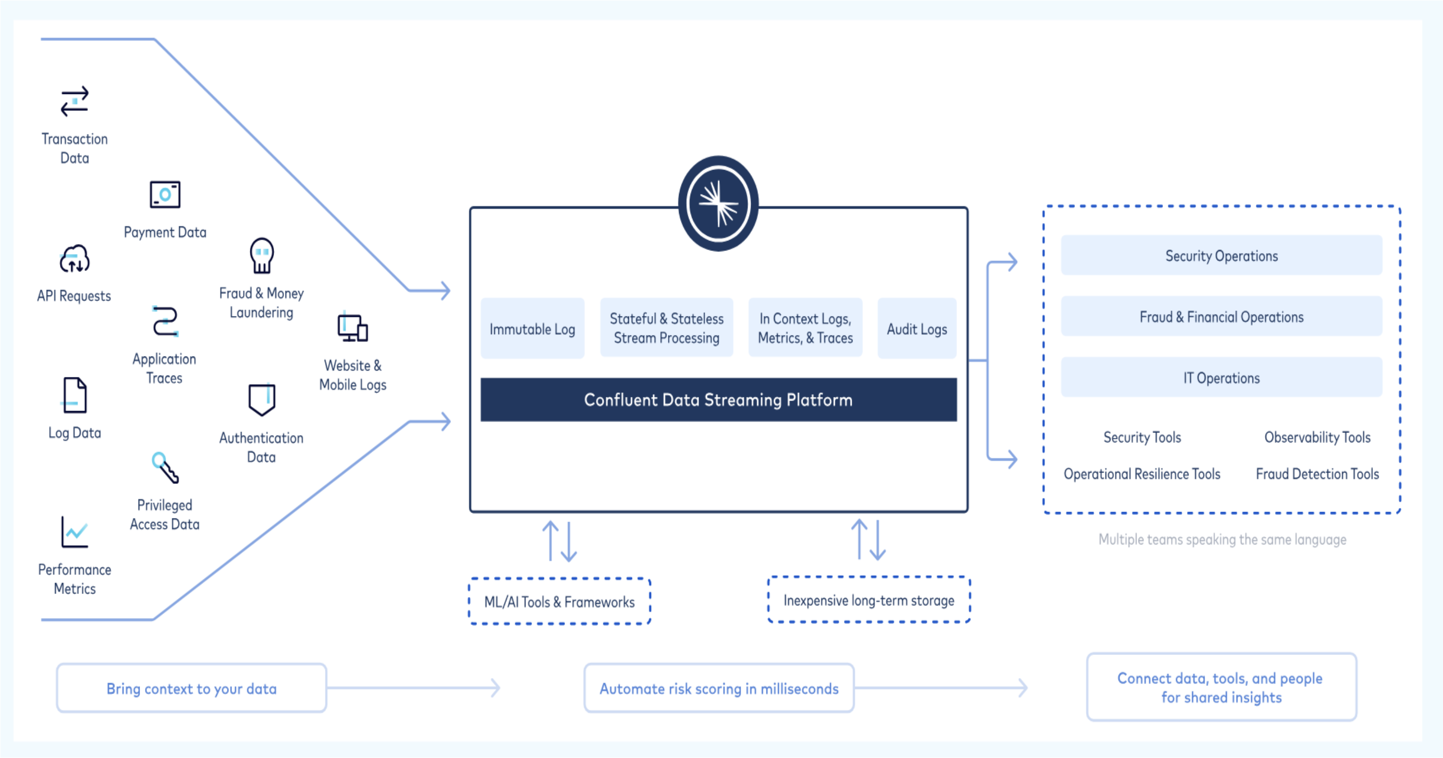
💰 $32B in global credit card fraud losses annually
⏰ Traditional systems: 2-4 hours detection delay
💸 Average fraud cost: $190 per transaction
❌ 10-15% false positive rate

Our Solution:

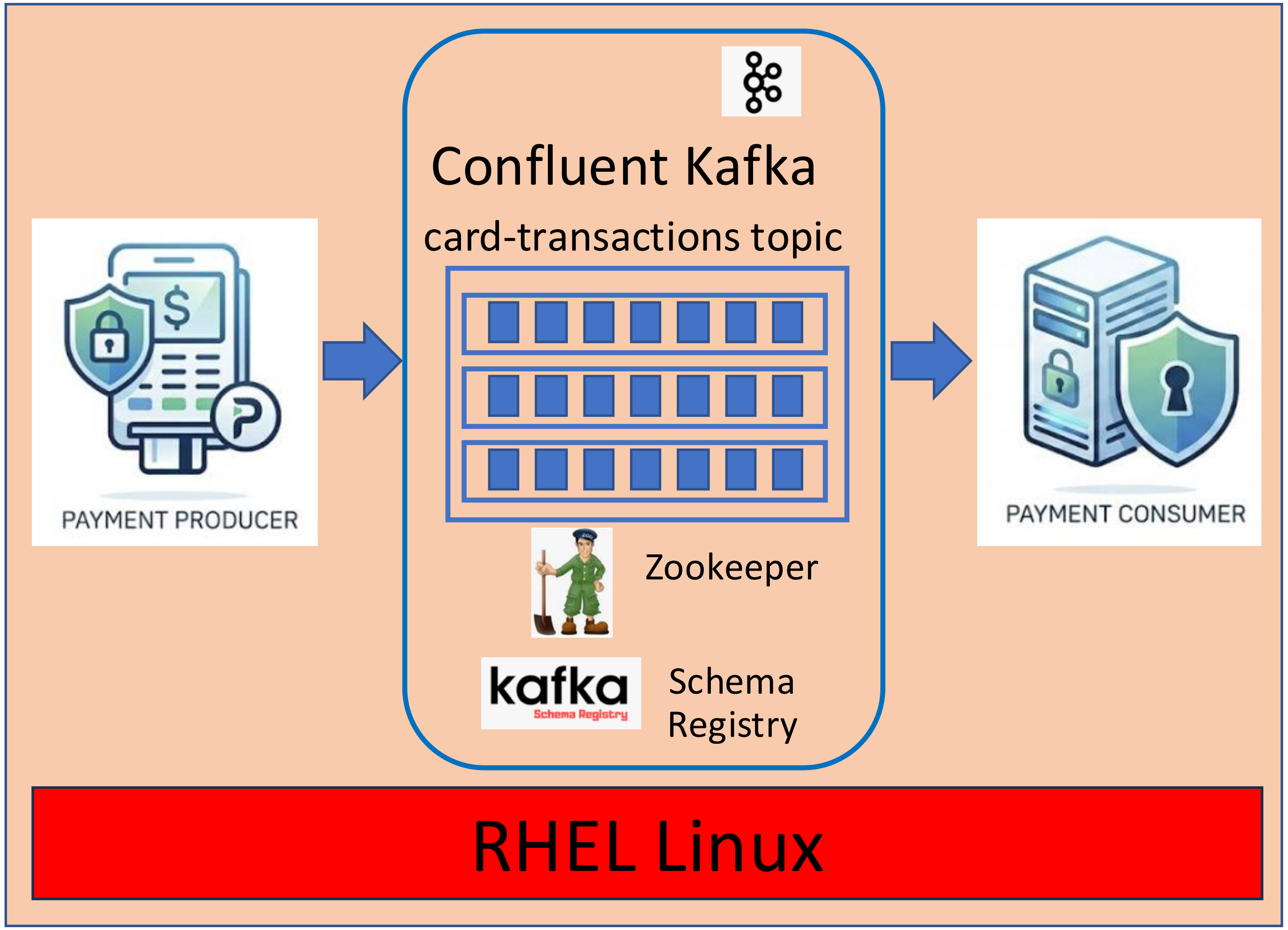
⚡ Fraud detection with <100ms latency
🔍 7 intelligent fraud detection patterns
📊 95% fraud detection accuracy
💰 $ 340K annual cost savings

Technology Stack:

Confluent Kafka Community Edition
Python 3.9+ with confluent-kafka library
Red Hat Enterprise Linux



System Architecture



Key Components:

Payment Producer (card_transaction_producer.py)
- Generates realistic transactions
- Embeds fraud patterns naturally (~20%)
- 1 transaction/second rate

Kafka Cluster
- Topic: card-transactions
- 3 partitions for scalability
- 7-day retention policy
- LZ4 compression

ZooKeeper and Schema Registry
- Manages Kafka cluster coordination, leader election, and configuration metadata
- Ensures data schema consistency and evolution across producers and consumers

Payment Consumer (card_transaction_consumer.py)
- Real-time fraud detection engine
- 7 pattern analysis algorithms
- Dynamic user profiling
- Risk scoring (0.0-1.0)

Fraud Detection Intelligence – developed by IBM Bob

How It Works:
Build user profiles dynamically (spending patterns, locations)
Analyze each transaction against 7 fraud patterns
Calculate cumulative risk score (0.0 - 1.0)
Flag transactions with risk ≥ 0.5 as fraudulent

Pattern	Risk Weight	Example
1. Unusually High Amount	+0.3	$1,200vs$80 avg
2. Absolute High Amount	+0.2	Any transaction >$300
3. Rapid Succession	+0.25	3+ txns in 2 minutes
4. High-Risk Location	+0.35	Nigeria, Russia, China
5. Suspicious Merchant	+0.3	Crypto exchanges, gift cards
6. Location Anomaly	+0.25	Unusual foreign location
7. Multiple Locations	+0.2	3+ cities in 1 hour

Risk Score Example:
High amount: +0.3
High-risk country: +0.35
Suspicious merchant: +0.3
Total: 0.95 → FRAUD! 🚨

Demo Results & Business Value

Live Demo Highlights:

Producer Output:
📤 Transaction #15
ID: TXN456789
User: USER1003
Merchant: Crypto Exchange
Amount: $850.00
Location: Lagos, Nigeria

Consumer Detection:
🚨 FRAUDULENT TRANSACTION DETECTED 🚨
Risk Score: 0.85
Fraud Indicators:
- High transaction amount ($850.00)
- High-risk country: Nigeria
- Suspicious merchant: Crypto Exchange

Business Impact

Demo Scale Assumptions:
Transaction Volume:
1,000 transactions/day (vs 100,000/day enterprise)
~15% fraud rate = 150 fraudulent transactions/day
Average fraud amount: $400

Cost Avoidance:
Prevented fraud: $400 \times 150 \times 365 =$219K/year
Reduced manual review: $50K/year
Infrastructure efficiency: $30K/year
Faster detection (vs batch): $41K/year
Total Annual Benefit: $340K

Link to solution on Github

https://github.ibm.com/vivprata/ConfluentCreditCardFraud/blob/master/CASE_STUDY.md